



MISE EN PLACE D'UN LABORATOIRE DE DÉTECTION D'INTRUSION AVEC SELKS ET SIMULATION D'ATTAQUES RÉSEAU

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SOMMAIRE

I)

INTRODUCTION

II)

surveillance le trafic réseau

III)

détection les alertes simples

IV)

Conclusion

Introduction

Face à la multiplication des cyberattaques, la détection d'intrusion est devenue indispensable pour protéger les systèmes d'information. Ce projet consiste à mettre en place un laboratoire virtuel sous VMware permettant de simuler et détecter des attaques réseau en temps réel. L'environnement repose sur trois machines virtuelles : Kali Linux comme machine attaquante, SELKS/ClearNDR comme système de détection (IDS), et Ubuntu 22.04 comme machine cible. Les outils utilisés incluent Nmap, Suricata et Wireshark.

L'objectif est de comprendre concrètement le fonctionnement d'un système de détection d'intrusion à travers des attaques simulées et l'analyse des alertes générées

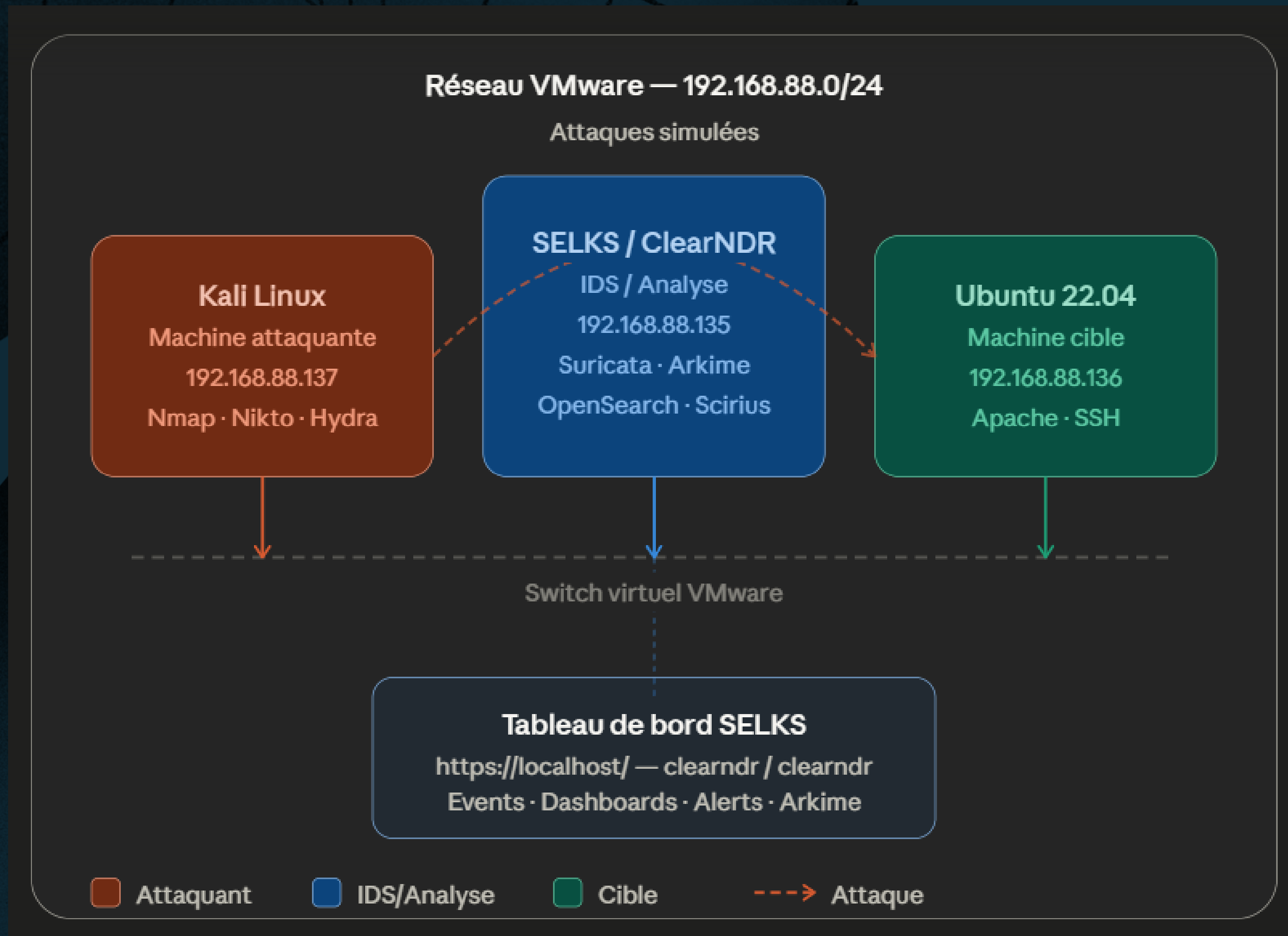
MACHINE VIRTUELLES NÉCESSAIRES

machine	role	IP
Kali linux	machine attaquante	192.168.88.137
ubuntu	machine cible	192.168.88.136
SELKS	IDS/analyse	192.168.88.132

outils utiliser

outils	fonction
SELKS	détection IDS
suricata	analyse réseau
Kali linux	simulation d'attaques
NMAP	scan réseau
Wirshark	analyse paquets

architecture



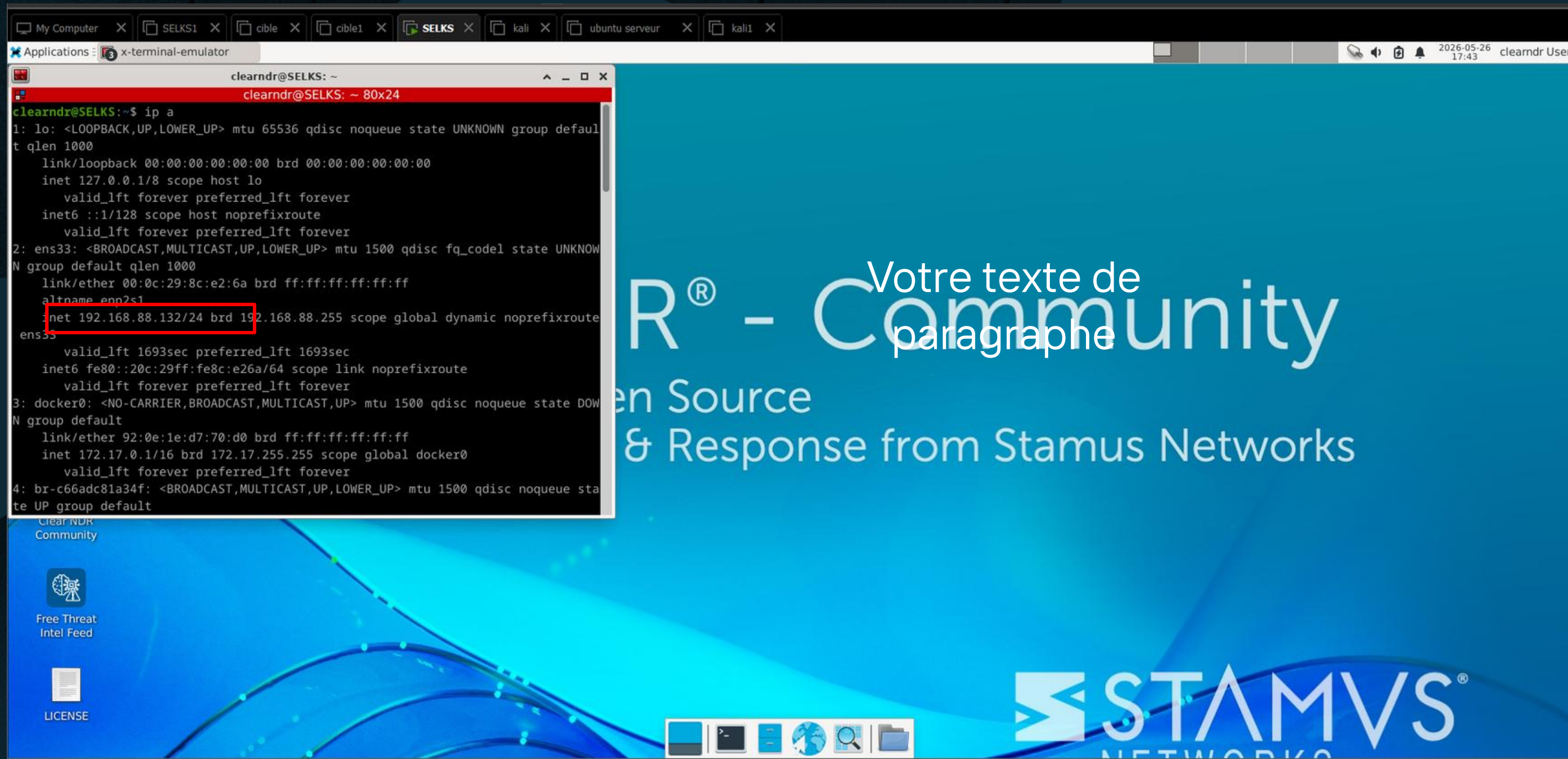
la machine cible(ubuntu)

```
cible@cible-virtual-machine:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 00:0c:29:6e:7d:c5 brd ff:ff:ff:ff:ff:ff
    altnames enp2s1
    inet 192.168.88.136/24 brd 192.168.88.255 scope global dynamic noprefixroute ens33
        valid_lft 1689sec preferred_lft 1689sec
    inet6 fe80::2da4:aacb:87d3:cc36/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
cible@cible-virtual-machine:~$ ping 192.168.88.137
PING 192.168.88.137 (192.168.88.137) 56(84) bytes of data.
64 bytes from 192.168.88.137: icmp_seq=1 ttl=64 time=1.23 ms
64 bytes from 192.168.88.137: icmp_seq=2 ttl=64 time=0.711 ms
```


la machine attaquante(kali)

```
kane@192: ~  
  
(kane@192) - [~]  
$ ping 192.168.88.135  
PING 192.168.88.135 (192.168.88.135) 56(84) bytes of data.  
64 bytes from 192.168.88.135: icmp_seq=1 ttl=64 time=1.03 ms  
64 bytes from 192.168.88.135: icmp_seq=2 ttl=64 time=0.341 ms  
64 bytes from 192.168.88.135: icmp_seq=3 ttl=64 time=0.779 ms  
^C  
--- 192.168.88.135 ping statistics ---  
3 packets transmitted, 3 received, 0% packet loss, time 2024ms  
rtt min/avg/max/mdev = 0.341/0.718/1.034/0.286 ms  
  
(kane@192) - [~]  
$ ip a  
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000  
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00  
    inet 127.0.0.1/8 scope host lo  
        valid_lft forever preferred_lft forever  
    inet6 ::1/128 scope host noprefixroute  
        valid_lft forever preferred_lft forever  
2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000  
    link/ether 00:0c:29:13:44:18 brd ff:ff:ff:ff:ff:ff  
    inet 192.168.88.137/24 brd 192.168.88.255 scope global dynamic noprefixroute
```

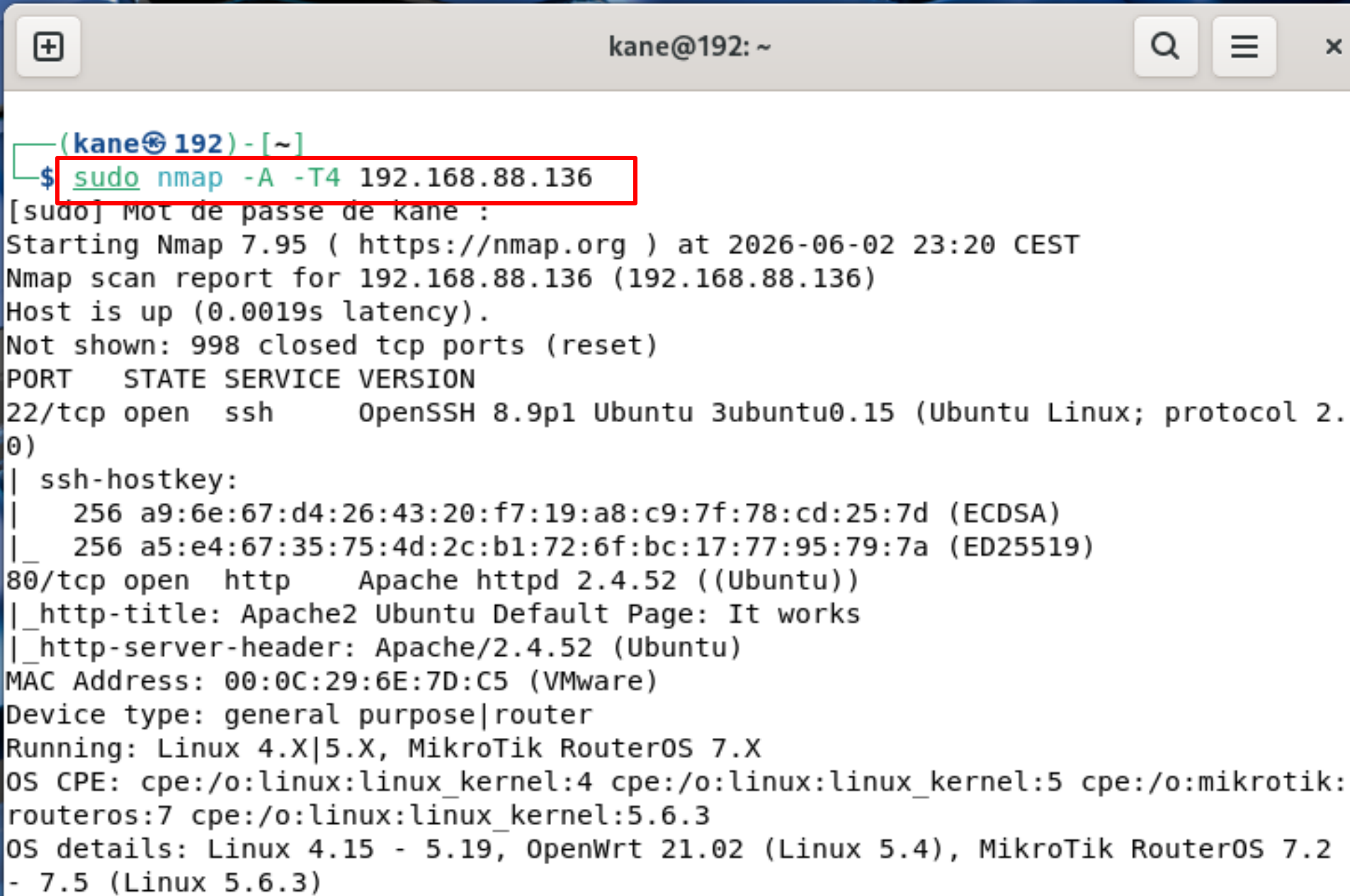

la machine SELKS



Le teste : commande utiliser sur kali

```
(kane@192) - [~]  
$ nmap -Pn -sV -sC 192.168.88.136  
Starting Nmap 7.95 ( https://nmap.org ) at 2026-06-02 23:19 CEST  
Nmap scan report for 192.168.88.136 (192.168.88.136)  
Host is up (0.0019s latency).  
Not shown: 998 closed tcp ports (reset)  
PORT      STATE SERVICE VERSION  
22/tcp open  ssh      OpenSSH 8.9p1 Ubuntu 3ubuntu0.15 (Ubuntu Linux; protocol 2.0)  
| ssh-hostkey:  
|   256 a9:6e:67:d4:26:43:20:f7:19:a8:c9:7f:78:cd:25:7d (ECDSA)  
|_  256 a5:e4:67:35:75:4d:2c:b1:72:6f:bc:17:77:95:79:7a (ED25519)  
80/tcp open  http      Apache httpd 2.4.52 ((Ubuntu))  
|_ http-title: Apache2 Ubuntu Default Page: It works  
|_ http-server-header: Apache/2.4.52 (Ubuntu)  
MAC Address: 00:0C:29:6E:7D:C5 (VMware)  
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel  
  
Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .  
Nmap done: 1 IP address (1 host up) scanned in 9.83 seconds  
  
(kane@192) - [~]  
$ sudo nmap -A -T4 192.168.88.136
```


Le teste : commande utiliser sur kali



A terminal window titled 'kane@192: ~' with search, menu, and close icons. The prompt is '(kane@192) - [~]'. The command 'sudo nmap -A -T4 192.168.88.136' is entered and highlighted with a red box. The output shows the Nmap scan results for 192.168.88.136, including host status, open ports (ssh, http), and OS details.

```
(kane@192) - [~]
$ sudo nmap -A -T4 192.168.88.136
[sudo] Mot de passe de kane :
Starting Nmap 7.95 ( https://nmap.org ) at 2026-06-02 23:20 CEST
Nmap scan report for 192.168.88.136 (192.168.88.136)
Host is up (0.0019s latency).
Not shown: 998 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 8.9p1 Ubuntu 3ubuntu0.15 (Ubuntu Linux; protocol 2.0)
| ssh-hostkey:
|   256 a9:6e:67:d4:26:43:20:f7:19:a8:c9:7f:78:cd:25:7d (ECDSA)
|_  256 a5:e4:67:35:75:4d:2c:b1:72:6f:bc:17:77:95:79:7a (ED25519)
80/tcp    open  http     Apache httpd 2.4.52 ((Ubuntu))
|_ http-title: Apache2 Ubuntu Default Page: It works
|_ http-server-header: Apache/2.4.52 (Ubuntu)
MAC Address: 00:0C:29:6E:7D:C5 (VMware)
Device type: general purpose|router
Running: Linux 4.X|5.X, MikroTik RouterOS 7.X
OS CPE: cpe:/o:linux:linux_kernel:4 cpe:/o:linux:linux_kernel:5 cpe:/o:mikrotik:
routeros:7 cpe:/o:linux:linux_kernel:5.6.3
OS details: Linux 4.15 - 5.19, OpenWrt 21.02 (Linux 5.4), MikroTik RouterOS 7.2
- 7.5 (Linux 5.6.3)
```


resultat dans SELKS (SURICATA)

Stamus - Events

https://localhost/stamus/hunting/events

STAMVS NETWORKS

Reloadlast 1 hourPreview new UIHelpclearmr

HUNTING

DashboardEventsDetection MethodsPolicies

ADMINISTRATION

SuricataUsersSourcesRulesetsOtherHistoryExternal links

OTHER APPS

ArkimeOpenSearchEveBoxCyberChef

HUNTING > EVENTS

Filters

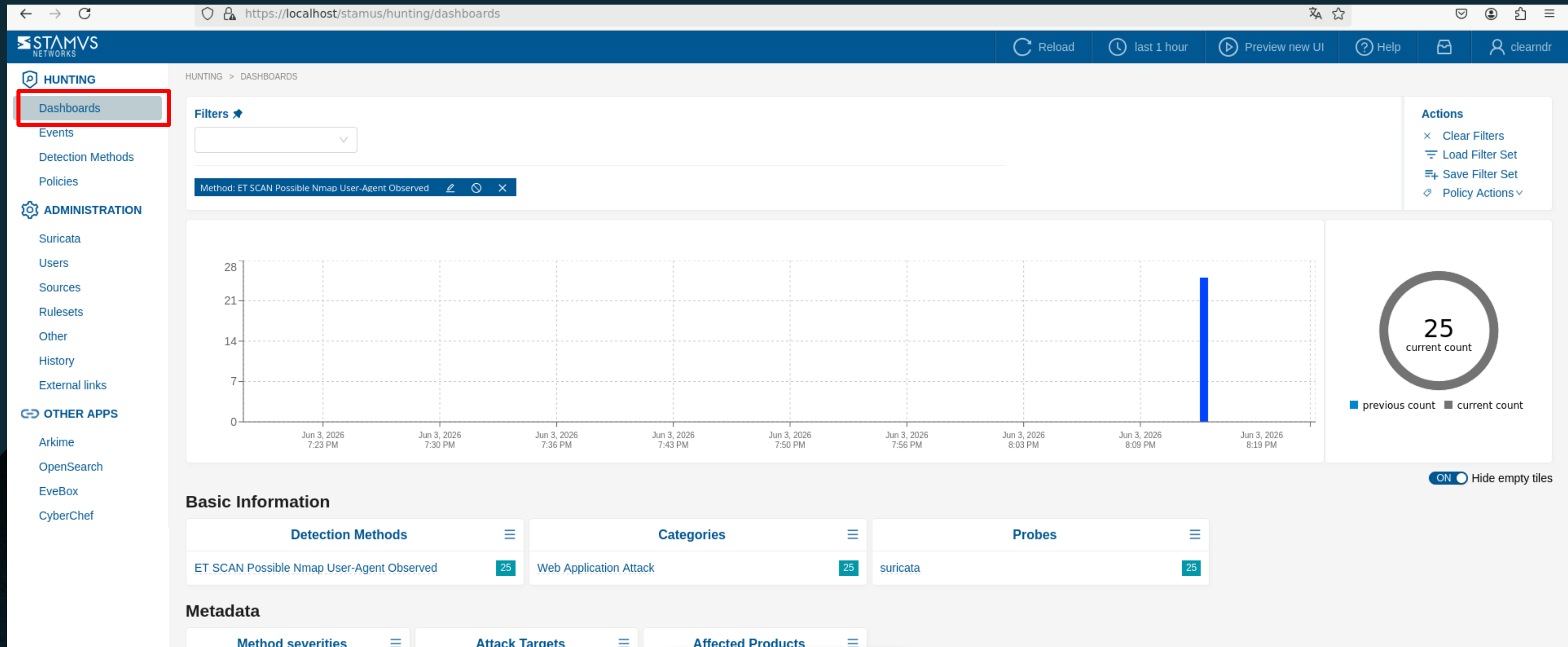
Actions

	Timestamp	Method	Source IP	Destination IP	Proto	Probe	Category	Tag
+ 🔔	2026-06-03, 08:13:53 pm	ET SCAN Possible Nmap User-Agent Observed	192.168.88.137	192.168.88.136	http	suricata	Web Application Attack	untagged
+ 🔔	2026-06-03, 08:13:53 pm	ET SCAN Possible Nmap User-Agent Observed	192.168.88.137	192.168.88.136	http	suricata	Web Application Attack	untagged
+ 🔔	2026-06-03, 08:13:53 pm	ET SCAN Possible Nmap User-Agent Observed	192.168.88.137	192.168.88.136	http	suricata	Web Application Attack	untagged
+ 🔔	2026-06-03, 08:13:53 pm	ET SCAN Possible Nmap User-Agent Observed	192.168.88.137	192.168.88.136	http	suricata	Web Application Attack	untagged
+ 🔔	2026-06-03, 08:13:53 pm	ET SCAN Possible Nmap User-Agent Observed	192.168.88.137	192.168.88.136	http	suricata	Web Application Attack	untagged
+ 🔔	2026-06-03, 08:13:53 pm	ET SCAN Possible Nmap User-Agent Observed	192.168.88.137	192.168.88.136	http	suricata	Web Application Attack	untagged
+ 🔔	2026-06-03, 08:13:53 pm	ET SCAN Possible Nmap User-Agent Observed	192.168.88.137	192.168.88.136	http	suricata	Web Application Attack	untagged
+ 🔔	2026-06-03, 08:13:53 pm	ET SCAN Possible Nmap User-Agent Observed	192.168.88.137	192.168.88.136	http	suricata	Web Application Attack	untagged
+ 🔔	2026-06-03, 08:13:53 pm	ET SCAN Possible Nmap User-Agent Observed	192.168.88.137	192.168.88.136	http	suricata	Web Application Attack	untagged
+ 🔔	2026-06-03, 08:13:53 pm	ET SCAN Possible Nmap User-Agent Observed	192.168.88.137	192.168.88.136	http	suricata	Web Application Attack	untagged

<123>

10 / page

resultat dans SELKS (SURICATA)



resultat dans SELKS (SURICATA)

Applications

Status - Events — Moz...

x-terminal-emulator

ApplicationsStatus - Events × +

← → ↺ https://localhost/stamus/hunting/events

STAMVS NETWORKS

Reloadlast 1 hourPreview new UIHelpclearndr

HUNTING > EVENTS

HUNTING

Dashboards

Events

Detection Methods

Policies

ADMINISTRATION

Suricata

Users

Sources

Rulesets

Other

History

External links

OTHER APPS

Arkime

OpenSearch

EveBox

CyberChef

Filters ★

Method: ET SCAN Possible Nmap User-Agent Observed

Actions

- Clear Filters
- Load Filter Set
- Save Filter Set
- Policy Actions ▾

Timestamp	Method	Source IP	Destination IP	Proto	Probe	Category	Tag
— 🔔 2026-06-03, 08:13:53 pm	ET SCAN Possible Nmap User-Agent Observed	192.168.88.137	192.168.88.136	http	suricata	Web Application Attack	untagged

Synthetic view

Related Alert 1

Related Flow 1

Related HTTP 1

JSON View

PCAP File

Detection Method

Detection Method

Method ET SCAN Possible Nmap User-...

Method ID 2024364

Category Web Application Attack

Severity Severe

Revision 5

IP and basic information

Source IP 192.168.88.137

Source port 34982

Destination IP 192.168.88.136

Destination port 80

IP protocol TCP

App protocol http

Probe suricata

Network Interface ens33

Enrichment

Offender IP 192.168.88.136

Offender port 80

Victim IP 192.168.88.137

Victim Port 34982

Flow

Flow start 2026-06-03T18:13:53.399923+0...

Client IP 192.168.88.137

Server IP 192.168.88.136

Bytes to server 487

Bytes to client 387

Pkts to server 4

Pkts to client 3

Flow ID 310285138534418

Community ID 1:gNaWYmqQq/BZZQoSakWsS...

Geo IP

Host 192.168.88.136

URL /

Status 200

HTTP

Host 192.168.88.136

URL /

Status 200

Ethernet

Source MAC 00:0c:29:13:44:18

Destination MAC 00:0c:29:6e:7d:c5

Detection method metadata

Affected Product Any

Attack Target Client_and_Server

Confidence Medium

To direct input to this VM, move the mouse pointer inside or press Ctrl+G.

resultat dans SELKS (wireshark)

Applications

*ens33

x-terminal-emulator

*ens33

FichierEditerVueAllerCaptureAnalyserStatistiquesTelephonieWirelessOutilsAide

Appliquer un filtre d'affichage ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
2662	758.255288783	192.168.88.137	192.168.88.136	TCP	66	35468 → 22 [ACK] Seq=1 Ack=43 Win=64256 Len=0 TSval=2461359775 TSecr=2853760689
2663	758.257839453	192.168.88.137	192.168.88.136	TCP	66	34974 → 80 [FIN, ACK] Seq=218 Ack=183 Win=64256 Len=0 TSval=2461359778 TSecr=2853760669
2664	758.258060130	192.168.88.136	192.168.88.137	TCP	66	80 → 34974 [ACK] Seq=183 Ack=219 Win=65024 Len=0 TSval=2853760692 TSecr=2461359778
2665	758.258207222	192.168.88.137	192.168.88.136	TCP	74	34982 → 80 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=2461359778 TSecr=0 WS=256
2666	758.258213695	192.168.88.136	192.168.88.137	TCP	74	80 → 34982 [SYN, ACK] Seq=0 Ack=1 Win=65160 Len=0 MSS=1460 SACK_PERM TSval=2853760693 TSecr=2461359778 WS=128
2667	758.258368735	192.168.88.137	192.168.88.136	SSHv2	93	Client: Protocol (SSH-2.0-Nmap-SSH2-Hostkey)
2668	758.258375007	192.168.88.137	192.168.88.136	TCP	66	34982 → 80 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=2461359779 TSecr=2853760693
2669	758.258376880	192.168.88.136	192.168.88.137	TCP	66	22 → 35468 [ACK] Seq=43 Ack=28 Win=65280 Len=0 TSval=2853760693 TSecr=2461359779
2670	758.260296749	192.168.88.137	192.168.88.136	HTTP	281	OPTIONS / HTTP/1.1
2671	758.260639501	192.168.88.136	192.168.88.137	TCP	66	80 → 34982 [ACK] Seq=1 Ack=216 Win=65024 Len=0 TSval=2853760695 TSecr=2461359780
2672	758.262402392	192.168.88.137	192.168.88.136	SSHv2	594	Client: Key Exchange Init
2673	758.262420396	192.168.88.136	192.168.88.137	SSHv2	1178	Server: Key Exchange Init
2674	758.262424854	192.168.88.136	192.168.88.137	HTTP	247	HTTP/1.1 200 OK
2675	758.262431026	192.168.88.137	192.168.88.136	TCP	66	34982 → 80 [ACK] Seq=216 Ack=182 Win=64256 Len=0 TSval=2461359782 TSecr=2853760696
2676	758.262433470	192.168.88.136	192.168.88.137	TCP	66	80 → 34982 [FIN, ACK] Seq=182 Ack=216 Win=65024 Len=0 TSval=2853760697 TSecr=2461359780
2677	758.265912137	192.168.88.136	192.168.88.137	TCP	66	22 → 35468 [FIN, ACK] Seq=1155 Ack=556 Win=64768 Len=0 TSval=2853760700 TSecr=2461359782
2678	758.266059927	192.168.88.137	192.168.88.136	TCP	66	35468 → 22 [ACK] Seq=556 Ack=1156 Win=65536 Len=0 TSval=2461359786 TSecr=2853760696
2679	758.268633180	192.168.88.137	192.168.88.136	TCP	66	34982 → 80 [FIN, ACK] Seq=216 Ack=183 Win=64256 Len=0 TSval=2461359788 TSecr=2853760697
2680	758.268650793	192.168.88.136	192.168.88.137	TCP	66	80 → 34982 [ACK] Seq=183 Ack=217 Win=65024 Len=0 TSval=2853760703 TSecr=2461359788
2681	758.270581224	192.168.88.137	192.168.88.136	TCP	74	35474 → 22 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=2461359791 TSecr=0 WS=256
2682	758.271046850	192.168.88.136	192.168.88.137	TCP	74	22 → 35474 [SYN, ACK] Seq=0 Ack=1 Win=65160 Len=0 MSS=1460 SACK_PERM TSval=2853760705 TSecr=2461359791 WS=128
2683	758.271278805	192.168.88.137	192.168.88.136	TCP	66	35474 → 22 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=2461359791 TSecr=2853760705

Frame 1: 62 bytes on wire (496 bits), 62 bytes captured (496 bits) on interface ens33, id 0

Ethernet II, Src: VMware_8c:e2:6a (00:0c:29:8c:e2:6a), Dst: IPv6mcast_02 (33:33:00:00:00:02)

Internet Protocol Version 6, Src: fe80::20c:29ff:fe8c:e26a, Dst: ff02::2

Internet Control Message Protocol v6

0000

33 33 00 00 00 02 00 0c 29 8c e2 6a 86 dd 60 06 33.....)..j..

0010

51 9c 00 08 3a ff fe 80 00 00 00 00 00 02 0c Q...:... ..

0020

29 ff fe 8c e2 6a ff 02 00 00 00 00 00 00 00)....j... ..

0030

00 00 00 00 00 02 85 00 70 34 00 00 00 00 p4....

resultat dans SELKS (wireshark)

My ComputerSELKS1ciblecible1SELKSkaliubuntu serveurkali1

Applications*ens33x-terminal-emulator

*ens33

FichierEditerVueAllerCaptureAnalyserStatistiquesTelephonieWirelessOutilsAide

Appliquer un filtre d'affichage ... <Ctrl-/>

No.	Time	Source	Destination	Protocol	Length	Info
2674	758.262424854	192.168.88.136	192.168.88.137	HTTP	247	HTTP/1.1 200 OK
2675	758.262431026	192.168.88.137	192.168.88.136	TCP	66	34982 → 80 [ACK] Seq=216 Ack=182 Win=64256 Len=0 TSval=2461359782 TSecr=2853760696
2676	758.262433470	192.168.88.136	192.168.88.137	TCP	66	80 → 34982 [FIN, ACK] Seq=182 Ack=216 Win=65024 Len=0 TSval=2853760697 TSecr=2461359780
2677	758.265912137	192.168.88.136	192.168.88.137	TCP	66	22 → 35468 [FIN, ACK] Seq=1155 Ack=556 Win=64768 Len=0 TSval=2853760700 TSecr=2461359782
2678	758.266059927	192.168.88.137	192.168.88.136	TCP	66	35468 → 22 [ACK] Seq=556 Ack=1156 Win=65536 Len=0 TSval=2461359786 TSecr=2853760696
2679	758.268633180	192.168.88.137	192.168.88.136	TCP	66	34982 → 80 [FIN, ACK] Seq=216 Ack=183 Win=64256 Len=0 TSval=2461359788 TSecr=2853760697
2680	758.268650793	192.168.88.136	192.168.88.137	TCP	66	80 → 34982 [ACK] Seq=183 Ack=217 Win=65024 Len=0 TSval=2853760703 TSecr=2461359788
2681	758.270581224	192.168.88.137	192.168.88.136	TCP	74	35474 → 22 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=2461359791 TSecr=0 WS=256
2682	758.271046850	192.168.88.136	192.168.88.137	TCP	74	22 → 35474 [SYN, ACK] Seq=0 Ack=1 Win=65160 Len=0 MSS=1460 SACK_PERM TSval=2853760705 TSecr=2461359791 WS=128
2683	758.271278805	192.168.88.137	192.168.88.136	TCP	66	35474 → 22 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=2461359791 TSecr=2853760705
2684	758.289677855	192.168.88.136	192.168.88.137	SSHv2	108	Server: Protocol (SSH-2.0-OpenSSH_8.9p1 Ubuntu-3ubuntu0.15)
2685	758.289845727	192.168.88.137	192.168.88.136	TCP	66	35474 → 22 [ACK] Seq=1 Ack=43 Win=64256 Len=0 TSval=2461359810 TSecr=2853760724
2686	758.292041856	192.168.88.137	192.168.88.136	SSHv2	93	Client: Protocol (SSH-2.0-Nmap-SSH2-Hostkey)
2687	758.292546361	192.168.88.136	192.168.88.137	TCP	66	22 → 35474 [ACK] Seq=43 Ack=28 Win=65280 Len=0 TSval=2853760727 TSecr=2461359812
2688	758.293846999	192.168.88.137	192.168.88.136	TCP	66	35468 → 22 [FIN, ACK] Seq=556 Ack=1156 Win=65536 Len=0 TSval=2461359814 TSecr=2853760696
2689	758.295393134	192.168.88.137	192.168.88.136	SSHv2	594	Client: Key Exchange Init
2690	758.297757676	192.168.88.136	192.168.88.137	TCP	66	22 → 35468 [ACK] Seq=1156 Ack=557 Win=64768 Len=0 TSval=2853760732 TSecr=2461359814
2691	758.297775579	192.168.88.136	192.168.88.137	TCP	66	22 → 35474 [ACK] Seq=43 Ack=556 Win=64768 Len=0 TSval=2853760732 TSecr=2461359816
2692	758.300699312	192.168.88.136	192.168.88.137	SSHv2	1178	Server: Key Exchange Init
2693	758.304959220	192.168.88.136	192.168.88.137	TCP	66	22 → 35474 [FIN, ACK] Seq=1155 Ack=556 Win=64768 Len=0 TSval=2853760739 TSecr=2461359816
2694	758.304976532	192.168.88.137	192.168.88.136	TCP	74	35488 → 22 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM TSval=2461359825 TSecr=0 WS=256
2695	758.304980831	192.168.88.137	192.168.88.136	TCP	66	35474 → 22 [ACK] Seq=556 Ack=1156 Win=65536 Len=0 TSval=2461359825 TSecr=2853760735

▶ Frame 1: 62 bytes on wire (496 bits), 62 bytes captured (496 bits) on interface ens33, id 0

▶ Ethernet II, Src: VMware_8c:e2:6a (00:0c:29:8c:e2:6a), Dst: IPv6mcast_02 (33:33:00:00:00:02)

▶ Internet Protocol Version 6, Src: fe80::20c:29ff:fe8c:e26a, Dst: ff02::2

▶ Internet Control Message Protocol v6

0000

33 33 00 00 00 02 00 0c 29 8c e2 6a 86 dd 60 06 33.....)..j..

0010

51 9c 00 08 3a ff fe 80 00 00 00 00 00 00 02 0c 0.....Q.....

0020

29 ff fe 8c e2 6a ff 02 00 00 00 00 00 00 00 00).j.....

0030

00 00 00 00 00 02 85 00 70 34 00 00 00 00 00p4.....

Conclusion

Ce projet nous a permis de mettre en place un laboratoire de détection d'intrusion fonctionnel, allant de l'installation des machines virtuelles jusqu'à la simulation et la détection d'attaques réseau en temps réel.

À travers cet environnement, nous avons pu constater l'efficacité de SELKS/ClearNDR et de son moteur Suricata dans la détection d'activités malveillantes telles que les scans de ports et les tentatives de connexion par force brute. Les alertes générées ont permis d'identifier précisément la source, la nature et la cible de chaque attaque.

Ce laboratoire constitue une base solide pour approfondir les compétences en cybersécurité défensive et explorer des scénarios d'attaques plus complexes.